

Taxi Cab Research

Taxi Cabs vs Rideshare has been a hot button topic within NYC with the NYC government enacting special laws to protect taxi cab drivers following the creation of rideshare apps such as Lyft and Uber.

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Version:	1.0
Target Community:	Both Taxi Cab Drivers and Taxi Cab Riders
Date Created:	09/10/2026
Last Updated:	09/18/2026
GitHub Repository:	https://github.com/colin-seiler/nyc-taxi-cab

1 Research Goal

Currently our current research goal is to explore both yellow and green taxi data alongside Uber and Lyft rideshare data. Our goal is to compare availability and cost of both systems. The idea started as trying to predict fare price of taxis but is being extended as an idea to try and predict price of taxis, ubers, and lyfts and provide a recommendation on that. There is no absolute ground truth, but we will build predictive models for each and compare their predictions.

2 Background and Motivation

Since the induction of Uber and Lyft, their impact on the hallmark taxi service in New York has been a hot button topic for citizens, requiring NYC to enact laws regarding the legality and licensing of drivers that drive for Lyft and Uber. NYC has a major focus on this due to the cabs being a signature look in the streets of manhattan and because of the high investment required to acquire a taxi cab medallion from the city. At one point, taxi cab medallions in NYC ran at around \$1-1.2 million. The price of these medallions crashed with the release of Lyft and Uber, now being priced around \$90-200k. The price of the taxi cab medallions should generally reflect the expected monetary value you can expect to make over a certain amount of time, we can explore if the value of a medallion is still considered worth it or if the cheaper entry fee of uber and lyft will be more worth it.

Beyond that, while NYC has a very impressive and useful subway system, taxis and rideshare options still generate an impressive amount of money. Exploring hot areas for each and finding which service is better in certain areas or for certain types of rides will be interesting.

3 Research Objectives and Scope

Our current objectives are open to change but as an outline we have 3 main objectives:

- Examine a few hot button areas, specifically around MSG, Penn Station, and the 3 Airports in NYC (LGA, JFK, EWR) and see which is service is used most and when.
- Examine by area in Manhattan (LES, FiDi, Midtown, Chelsea, etc) and some popular Bronx/Queens areas (Williamsburg, Greenpoint, Astoria, Dumbo, etc) and see the popularity of both and general costs.
- Build a predictive tool that generates predicted fare cost of all 3 services as a comparison

We are open to exploring our data in depth but we are not going to attempt to examine how ride share has impacted taxi prices but instead look to explore and evaluate the current value of each as seperate entities.

4 Prior Research and References

Medium - NYC Taxi Trip Data Analysis

This article and work is a great outline of the initial steps we can take to perform cleaning and eda. It talks through all the different data quality issues we will run into.

Data Visualization of NYC Taxi Trip Data

This article is good as a guide, outlining and giving us a good look at the data before we dig into it ourselves

Rideshare Comparison Shop

Article that uses a seperate dataset but builds process to examine how well customers generally shop around or even care to compare.

Ride Sharing Case Study

Great article that compares how things like surging and other features influence costs

Fare Comparison App of Uber, Ola and Rapido
An article that outlines the same goal that we are trying to investigate

5 Supporting Data and Resources

We expect to add a few other datasets to our source material in the future. We expect to cross examine versus important events at some point.

As for software and APIs, we currently are aiming to use databricks for working with our data and cleaning and processing it. We will attempt to build through the entire medallion architecture. Our source material, again, will be the NYC Taxi Cab Open source data.

6 Risks, Constraints, Assumptions, and Open Questions

The data is all available and easily accesible so we expect no risks in regards to this. The data is incredibly large, so it may take some large amount of time to process. One small risk is we would be building 3 seperate models on Lyft, Uber, and Taxis. In that sense we don't have any ground truth where we would see actual comparisons across the 3 in real time and a choice being made. That is a risk in our ability to analyze the data.

7 Research Approach

- Week 1: Understand the data, begin cleaning and processing in a simple manner. Bridge all schemas in case of quality issues
- Week 2: Begin building ways to automate the retrieval and processing of all the data in databricks
- Week 3: In medallion architecture terms, we want to have our data all set in gold tables and clean by the middle of this week, having made a lot of decision on our data, and build out some analysis and EDA
- Week 4: Finally, at this point we should have our analysis complete and ready for presentation in a longer presentation session for phase 2 as outlined
- Following Weeks: After verifying our data is processed and analyzed properly we will begin building out the models to verify and compare across Taxi, Uber, Lyft. I do wonder if this will be hard enough, I'm sure we will find some further research to explore.